Zero-Shot Monocular Depth Estimation for Metric Object Sizing in Low-Contrast Environments

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***Abstract*—We ask how far a purely zero-shot pipeline can go from a single RGB image to tape-measured object dimensions. The pipeline pairs a frozen YOLOv8m detector with a frozen monocular depth model and converts relative depth into metres with a two-anchor disparity fit; a pinhole back-projection then yields each object's distance Z, width W and height H. On a purpose-built 100-image dataset (five household objects, five distances from 0.50 to 0.90 m, four lighting conditions including an active-infrared night mode) we benchmark five depth backbones: Depth Anything V2 Small/Base/Large, MiDaS (DPT-Hybrid) and Marigold. Depth Anything V2 Large is best, with 3.63 cm mean absolute distance error (4.9 %), 9.3 % / 4.9 % relative width / height error and no gross failures, against 9.38 cm for MiDaS. The aggregate hides a strongly non-uniform error profile: 61 % of the total absolute distance error comes from one tall, thin, clear bottle, and that error is a one-directional over-estimation almost entirely explained by where depth is sampled inside the box; sampling the ground-contact point instead of the box interior cuts it by 40 % at no cost. Night-time infrared capture is the second failure mode, and its error tracks highlight clipping rather than low contrast. We also show that 5 % of detection rows must be voided rather than scored, and that scoring them silently can roughly double a model's reported error.**

***Index Terms*—monocular depth estimation, zero-shot inference, object sizing, YOLO, geometric calibration, benchmark**

I. Introduction

Turning a photograph into a physical measurement recurs in agriculture, logistics, inventory and robotic manipulation: a system that reports "this fruit is 7.4 cm across and 62 cm away" from the camera it already has needs no depth sensor, no stereo rig and no calibration target on the object. Monocular depth estimation (MDE) is strong in the zero-shot regime [2], [5], but published benchmarks report relative-depth metrics on internet-scale imagery, not centimetres on a tableside object under a dim bulb. This paper asks: *if we freeze a detector and a depth model and bolt a two-anchor calibration onto them, how accurate are the resulting metric dimensions, and where does that accuracy break?*

We answer with a 100-image capture matrix that isolates object identity, distance and illumination on a locked-down rig, with every distance and object dimension tape-measured. Every model is used as shipped: no fine-tuning, no per-scene optimisation, no test-time adaptation. Our contributions are:

- a reproducible zero-shot pipeline that converts detector boxes plus relative depth into per-object Z, W and H (Section III), with a 100-image dataset spanning lighting, distance and object axes;
- a five-model benchmark on that dataset, reported in aggregate and decomposed by lighting, distance and object (Section V);
- the finding that the aggregate error is dominated by a single object geometry, that this error is a signed bias rather than noise, and that a one-line change to depth sampling removes 40 % of it (Section V-F);
- an evaluation-integrity result: without a detection gate, 10 % of rows silently size the wrong object and inflate reported error by up to 2× (Section V-G).

II. Related Work

Eigen *et al.* [1] established single-image depth regression; MiDaS [2], [3] scaled it and made cross-dataset zero-shot transfer the norm. Depth Anything V1/V2 [4], [5] extend that recipe with large-scale unlabelled data, while Marigold [6] re-casts depth as conditional diffusion on a pretrained image generator. All of these predict *relative* or affine-invariant depth, so calibration is unavoidable for metrology; metric variants (ZoeDepth [7], Metric3D [8]) are tied to the camera and domain of their training data, which is what a plug-and-play deployment avoids.

We take object extents from YOLO [9] in its Ultralytics form [10]. Boxes are the cheapest localisation available and their tightness becomes a first-order error term for thin objects (Section V). Back-projecting pixels through a pinhole model is textbook geometry [13]; the open question is the depth stage feeding it. Existing benchmarks (KITTI [11], NYU-Depth [12]) use LiDAR or RGB-D ground truth in driving or room scenes and report scale-invariant errors, so they do not answer how a commodity camera with an assumed field of view performs on small objects at 0.5–0.9 m under warm, dim and infrared light.

III. Method

A. Overview

Figure 1 shows the five stages. A 1920×1080 RGB frame goes to YOLOv8m, which returns boxes, classes and confidences, while a frozen depth model returns a dense relative-depth map D. Each surviving box selects a scalar depth statistic, the anchors calibrate it to metres, and the pinhole model turns the box extent at that depth into physical width and height, one row per object.

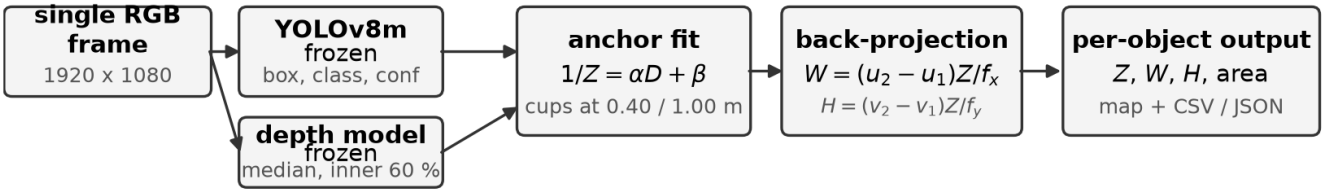


Fig. 1. The pipeline. Both learned components are frozen. Detections are gated to the expected class, the depth map is sampled inside each surviving box, and a two-anchor least-squares fit in disparity space maps relative depth to metres before the pinhole back-projection produces Z, W and H.

B. Disparity-Domain Anchor Calibration

Let $D(u, v)$ be the raw model output. The natural alignment is linear in depth,

$$Z(u, v) = \alpha\, D(u, v) + \beta, \tag{1}$$

with (α, β) fitted by least squares to tape-measured distances. Relative-depth models, however, are trained against *inverse* depth, so D behaves like disparity and fitting (1) warps any object lying between the calibration anchors. We instead fit

$$\frac{1}{Z(u, v)} = \alpha\, D(u, v) + \beta \quad \Longrightarrow \quad Z(u, v) = \frac{1}{\alpha\, D(u, v) + \beta}, \tag{2}$$

which is exact for a model whose output is proportional to disparity and interpolates correctly between anchors. Table I isolates the choice: on 25 white-light frames (five objects at 0.50–0.90 m) the two forms see identical boxes, anchors, detector and depth sampling, and only the fitted curve differs. For the four regression heads the linear form is not merely worse, it fails in a structured way: its mean signed error equals its MAE row for row, i.e. *every* prediction is long, because a straight line pinned to two inverse-depth anchors lies above the true curve everywhere between them. The penalty is 2.7–3.6× in MAE for the Depth Anything V2 sizes (Large: 3.42 to 12.42 cm) and 1.35× for MiDaS. Marigold is the sole exception (6.32 against 6.39 cm), which is what its affine-invariant, direct-depth output predicts; we therefore keep it under the same fit and flag the parity in Section VI rather than re-tuning one model. All results below use (2), which is also the default of the implementation.

Two physical reference cups are placed at known distances, 0.40 m and 1.00 m, in every scene and clicked once per scene folder, so calibration costs two clicks and no per-object measurement. The anchors bracket the working range (0.50–0.90 m), so the object always lies between them.

C. Depth Sampling and Back-Projection

A box covers a surface patch, not a point, so a scalar depth must be chosen. Our default takes the median of D over the interior 60 % of the box, which is robust to stray background pixels; Section V-F shows this is the wrong choice for tall, thin objects and evaluates alternatives. Given Z at the object and focal lengths (f_x, f_y), the pinhole model gives

$$W = \frac{|u_2 - u_1|\, Z}{f_x}, \qquad H = \frac{|v_2 - v_1|\, Z}{f_y}, \tag{3}$$

for a box spanning $u_1..u_2$ and $v_1..v_2$. The stream carries no EXIF, and the datasheet publishes a focal length (4.0 mm) but no active sensor width, which is the quantity needed to convert it to pixels, $f_x = f_{\mathrm{mm}} \cdot 1920/w_{\mathrm{mm}}$; nor can the width be recovered from the field-of-view figures, which are mutually inconsistent: 75.2° horizontal with 40.8° vertical implies neither a 16:9 nor a 4:3 frame and a diagonal of 81.1° rather than the quoted 88.3°, and a 4.0 mm lens on the quoted 1/3-in sensor would give 61.9° horizontal ($f_x \approx 1600$ px). We therefore assume a 65° horizontal field of view, giving $f_x = f_y = 1506.9$ px. Back-solving f_x from the tape-measured widths of the three compact, opaque objects (275 of the 475 scored rows) gives a median of 1478 px, 1.9 % below the assumed value, so the assumption is consistent with the data to within the box and depth noise. Note that f_x enters only Eq. (3): it scales every W and H by $1/f_x$ and leaves Z untouched, because Z is fitted from the anchors. The assumption is revisited in Section VI.

TABLE I
CALIBRATION-FORM ABLATION ON THE WHITE-LIGHT SET (N=25 PER CELL, FIVE OBJECTS, 0.50–0.90 M, YOLOV8M AT 0.25, BOX-MEDIAN SAMPLING). AF: LINEAR IN DEPTH (1); INV: LINEAR IN DISPARITY (2). EVERY AFFINE ROW HAS BIAS = MAE, I.E. ALL OF ITS ERRORS SHARE ONE SIGN.

	AF (1)		INV (2)		
Model	MAE (cm)	rel (%)	MAE (cm)	rel (%)	AF/INV MAE
DA-V2-S	12.16	18.1	**4.45**	**4.5**	2.73
DA-V2-B	13.29	20.1	**4.73**	**6.0**	2.81
DA-V2-L	12.42	18.7	**3.42**	**4.2**	3.63
MiDaS	16.58	25.5	**12.32**	**15.8**	1.35
Marigold	**6.32**	**7.0**	6.39	7.0	0.99

D. Detection Gating and Row Voiding

An early version scored whatever detection was most confident between the two anchors. Since the anchors are

themselves cups, that frequently scored a *calibration cup*, a knife or the table, and because ground-truth distance comes from the filename the resulting "errors" were 10–50 cm of pure bookkeeping (Section V-G). The pipeline now applies three rules: (i) drop any detection whose box contains an anchor click, which is exact because the clicks land on the anchor cups; (ii) keep only detections of the class the scene is about, from the object id or a target flag; (iii) if none survives, void the row instead of scoring it. Voided rows are logged, excluded from every metric, and counted in the summary.

IV. Experimental Setup

A. Dataset and Rig

The dataset contains 100 images: five objects $\times$ five distances (0.50 to 0.90 m in 10 cm steps) $\times$ four lighting conditions. The objects, with tape-measured $W \times H$, are a bowl (12.0×6.0 cm), a mug (12.0×8.0 cm), an apple (7.5×7.0 cm), a short, blue, translucent bottle (7.5×25.0 cm) and a tall, narrow, clear bottle (6.2×24.0 cm). The lighting modes are *white*, evenly lit and used as reference; *warm*, a strong yellow cast; *dim*, low and flat; and *nv*, lights off with an active-infrared illuminator, producing a near-monochrome frame. The camera is a TP-Link Tapo C200 (1/3-in progressive-scan CMOS, fixed 4.0 mm lens, $f/2.0$, 1920×1080 stream; vendor figures 88.3° diagonal / 75.2° horizontal / 40.8° vertical). It is locked on a tripod and no capture parameter changes. The two reference cups sit at 0.40 m and 1.00 m in every scene and the measured object is placed at the target distance with a tape measure. One image per cell, so the design measures systematic behaviour, not shot-to-shot variance.

B. Models and Implementation

We benchmark five frozen backbones: Depth Anything V2 Small, Base and Large (the `-hf` Transformers checkpoints) [5]; MiDaS in its DPT-Hybrid form [2], [3]; and Marigold [6]. Detection is YOLOv8m [10] at confidence 0.25 for every model. All five see identical boxes, anchors and calibration, with box-median depth sampling. Runs used one RTX 4070 (12 GB). Five models times 100 images gives 500 (model, image) pairs, of which 475 are scored and 25 voided.

C. Metrics

For each scored row we report $|\hat{Z} - Z|$ and $|\hat{Z} - Z|/Z$ and derive MAE, RMSE, mean signed bias and the fraction exceeding 20 cm or 30 %. Sizing accuracy is mean absolute relative error of W and of H against the tape-measured table.

V. Results and Discussion

A. Overall Accuracy

Depth Anything V2 Large wins every column of Table II: 3.63 cm MAE (4.9 %), 5.15 cm RMSE, 9.3 % width and 4.9 % height error, and no detection beyond 20 cm or 30 %. Its RMSE sits close to its MAE, so errors are small and evenly spread; Small's RMSE is nearly twice its MAE (10.71 vs 5.78 cm), meaning a few rows drive it, and that fragility rather than the mean is the real difference between the sizes.

TABLE II
Overall accuracy over 475 scored rows per model (25 voided rows excluded). Bias is the mean signed error; >20 cm and >30 % are gross-failure rates. Best value in each column is bold.

Model	MAE (cm)	rel (%)	bias (cm)	RMSE (cm)	>20 (cm)	>30 (%)	W (%)	H (%)
DA-V2-S	5.78	8.0	+4.77	10.71	12	7	12.9	8.3
DA-V2-B	5.35	7.4	+4.99	7.83	5	1	11.8	6.1
DA-V2-L	**3.63**	**4.9**	+3.21	**5.15**	**0**	**0**	**9.3**	**4.9**
MiDaS	9.38	13.1	+5.26	15.91	12	12	15.5	11.1
Marigold	8.55	12.1	−3.12	11.62	3	2	14.3	13.5

TABLE III
MAE of Z in cm / mean relative error in %, by lighting condition (rows 1–4) and by object (rows 5–9). The apple has $n = 15$ per model and the *nv* rows $n = 20$, because the apple is never detected at night. Marigold is omitted here and discussed in the text.

Condition	DA-V2-S	DA-V2-B	DA-V2-L	MiDaS
by lighting				
dim	4.47 / 6.1	4.66 / 6.3	**2.90 / 3.9**	5.42 / 8.0
white	4.45 / 6.1	4.73 / 6.5	3.42 / 4.5	12.32 / 17.1
warm	5.16 / 7.2	5.08 / 7.1	3.72 / 5.1	6.93 / 10.1
nv	9.84 / 13.9	7.33 / 10.3	4.69 / 6.6	13.73 / 18.3
by object				
bowl	**1.23 / 1.9**	1.73 / 2.3	1.67 / 2.2	7.13 / 9.0
mug	2.42 / 3.5	1.43 / 2.2	**1.21 / 1.8**	3.77 / 5.1
apple	**1.57 / 2.2**	2.95 / 3.9	2.37 / 3.1	4.63 / 6.7
bottle, short	**2.49 / 3.6**	6.17 / 8.4	3.73 / 4.8	20.23 / 29.0
bottle, tall	20.12 / 27.3	13.89 / 19.4	**8.86 / 12.4**	9.97 / 14.2

Base is a defensible middle ground. Accuracy is monotone in model size within the family, and the family beats both baselines on distance *and* sizing. MiDaS (9.38 cm) is weakest overall; Marigold lands at 8.55 cm at diffusion-sampling cost (Section VI).

Every model except Marigold over-estimates distance, with mean $\hat{Z}/Z$ of 1.062 (Small), 1.067 (Base), 1.042 (Large) and 1.075 (MiDaS); Marigold under-estimates at 0.949. One multiplicative correction fitted on the anchors would recover 4–7 points of relative error for free, and would improve W and H in proportion since both scale linearly with Z.

B. Effect of Lighting

Table III overturns the intuition that a dim scene is the hard case. Pooled over the Depth Anything family, dim is marginally the *easiest* condition at 4.01 cm, ahead of white (4.20) and warm (4.65). A plausible mechanism is figure/ground separation: darkening the frame leaves a bright object against a dark background, exactly the signal a depth model exploits. Marigold behaves differently: it is most accurate in white light (9.4 %) and worst under infrared (14.7 %), the opposite ordering to the Depth Anything family.

Night-time infrared is the failure mode: 7.29 cm pooled, 1.8× the dim error, and home to 14 of the 30 detections whose distance error exceeds 20 cm. Large degrades gracefully there (4.69 cm, no outliers), Small (9.84 cm) and MiDaS (13.73 cm)

do not. Measured luminance rules out low contrast: nv frames have the *highest* pixel standard deviation (52.3) and dim the lowest (21.2), yet nv is worst. The nv frame is high-contrast because it is clipped (p95 luminance 208.8, p5 28.8): blown highlights, crushed shadows, and the mid-tones carrying object texture removed.

C. Object Profile: What Is Actually Solved

The object panel of Table III is the most useful view of the results. Broad, squat, opaque objects are effectively solved: bowl and mug sit at 1.2–2.4 cm and under 4 % for every Depth Anything size, comfortably usable for sizing, and the apple is similarly easy *when detected* (1.57 cm for Small). The two bottles are the exception, and they are the only non-opaque objects: the tall one clear, the short one blue-tinted and translucent. Together they supply 78.7 % of all absolute distance error (pooled over the Depth Anything family): 61 % from the tall bottle and 17.7 % from the short one, against 21 % for the three compact, opaque objects.

This resolves an apparent paradox. Large wins overall, but on the four objects other than the tall bottle the ranking inverts: Small achieves 1.95 cm against Large's 2.23 cm, so Large's advantage is earned entirely on the tall bottle and at night. Since Small's weights are 99 MB against Large's 1.34 GB, a factor of 13, this is a deployment finding rather than a footnote: for near-range sizing of compact objects the smallest model suffices. MiDaS has an inverted object profile rather than a uniformly worse one: by far the worst on the short bottle (20.23 cm, 29 %) where Depth Anything stays at 2.5–6.2 cm, yet the least biased on the tall bottle.

D. Width and Height Accuracy

The deliverable is dimensions, not distance, and height is roughly twice as accurate as width for every model (Table II): it is the long axis of these objects, so a fixed box error is proportionally smaller, and vertical edges against a table surface are clean. Width error exceeds distance error on both bottles, which is a box-tightness signature rather than a depth signature. The short bottle shows 11.1 % distance error against 18.6 % width error, the tall bottle 18.0 % against 25.4 %: about 7.5 points on each come from box geometry no depth model can fix, so roughly 40 % of the bottle width error is box and 60 % is depth. The apple shows no such excess (+0.3 points), so its width error is entirely depth; height shows none anywhere.

E. Error Versus Distance

Figure 2 shows absolute error growing monotonically with distance for every model but Marigold. Relative error grows much more slowly, to about 5–7 % for Large, the signature of a correctly scaled inverse-depth fit: the geometry is sound and the limits are resolution, sampling and model capacity. The effective range is 0.50–0.80 m, where Large stays under 4.5 cm; nothing here supports sizing beyond 1 m without closer anchor spacing.

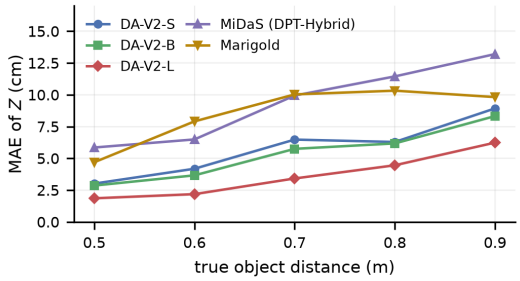


Fig. 2. Mean absolute distance error against true object distance, all objects. Absolute error grows monotonically for every model but Marigold, while relative error grows far more slowly (Section V), the signature of a correctly scaled inverse-depth fit.

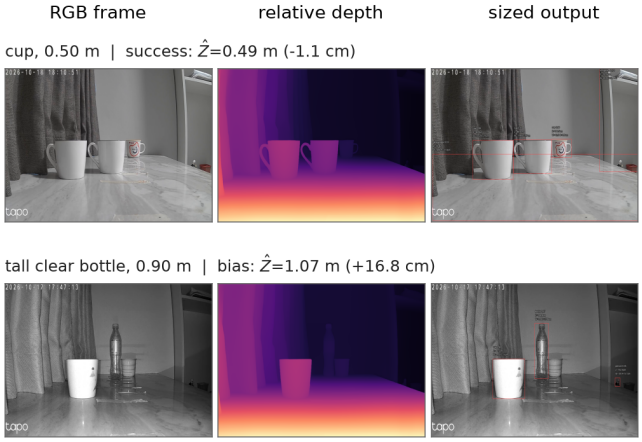


Fig. 3. Two frames with Depth Anything V2 Large: a mug at 0.50 m in white light (top) and the tall clear bottle at 0.90 m under infrared (bottom). Columns are the RGB input, the relative-depth map (magma, brighter = closer) and the sized output. The mug is sized to within 1.1 cm; the bottle, which barely separates from the background in the depth map, is read 16.8 cm too far, and its width error propagates into a reported 9.1 cm width for a 6.2 cm object.

F. The Tall Bottle: A Signed Bias, Not Noise

The tall bottle's error is almost entirely one-directional. For Large its MAE of 8.86 cm and its mean signed bias of $+8.86$ cm coincide, so essentially 100 % of the error is over-estimation, and the bias grows with distance, from $+5.34$ cm at 0.50 m to $+14.05$ cm at 0.90 m. Small is worse in the same direction, reading a 0.90 m bottle at roughly 1.32 m.

The cause is where depth is sampled, not the depth model; Figure 3 shows the case, where the bottle barely separates from the background in the depth map while the opaque mug pops out. The non-opaque material is a shared handicap, since the short bottle is translucent and still carries 17.7 % of the total error, but it does not set the magnitude: that bottle stays at 2.5–6.2 cm for the Depth Anything models, while the tall, colorless one reaches 8.9–20.1 cm. The difference is silhouette and box tightness. A 24 cm-tall, 6.2 cm-wide object inside a slightly loose box picks up background columns on both sides; background is farther, so a median over the interior drifts outward. The tall bottle is also the loosest detection in the

set, with confidence down to 0.67. A controlled sweep on the hardest folder confirms the diagnosis: on the night/tall-bottle images with the Small model and identical anchors, the box median gives 28.54 cm MAE (39.1 %), a box-centre sample 26.12 cm (35.5 %), and a bottom-centre sample 17.03 cm (23.0 %). Sampling the ground-contact point is a 40 % cut on the object carrying 61 % of the total error, at zero cost, and it is geometrically the right choice for a standing object because it is immune to loose horizontal box edges. (That sweep used the default YOLOv8 nano detector, so its absolute level is not comparable with Table II; the ordering is the result.)

G. Voided Rows and Evaluation Integrity

Of 500 (model, image) pairs, 25 are voided and all 25 are the apple at night: in every night frame of that folder YOLOv8m returns a cup, a knife, a bowl or a person, never an apple, so there is nothing to size and those rows are excluded rather than scored against a substituted object. This matters more than 5 % suggests. An earlier evaluation selected whichever detection sat most confidently between the anchors, with no class filter, and so scored calibration cups, knives, bowls and tables in 51 of 500 rows (10.2 %). Those rows averaged 39.5 cm error against 6.5 cm for correctly scored rows and supplied 60–90 % of each model's squared error; one row that sized a knife at 668 cm was 88 % of Marigold's total squared error. Correcting it roughly halved every model's RMSE and changed the object ranking. A pipeline with no expected-class gate produces plausible, entirely spurious metric errors, and a per-object breakdown is the cheapest defence.

VI. Limitations

Single camera, assumed intrinsics. The focal length comes from an assumed 65° horizontal field of view because the device provided no usable EXIF and its published field-of-view figures are mutually inconsistent. The assumption was checked against the data, not guessed: back-solving f_x from the objects' tape-measured dimensions gives 1478 px against the assumed 1506.9 px, so width and height may be systematically low by about 2 %. Real intrinsics would shift every absolute number, not the relative comparisons; feeding the vendor figures instead ($f_x = 1245$ or 1085 px) inflates W by 28–47 % on the same frames.

One shot per condition. A single image per cell does not separate model variance from scene variance.

Marigold is not reproducible as configured. The diffusion sampler starts without a fixed generator or latents and ran at 10 steps with one ensemble member at 768 px rather than 50 steps and 10 members; its number moved from 17.46 to 8.55 cm between runs, so it should not be quoted as a Marigold result until the seed and schedule are fixed. All five models, Marigold included, were evaluated under the same inverse-depth fit (2), so the comparison is like for like; Table I shows that for Marigold the two fits are within 0.1 cm of each other, so the shared fit costs it nothing measurable either way.

Night is confounded with clipping. Until an exposure bracket is captured, "depth degrades under infrared" is not separable from "this camera clipped the highlights under infrared". Tall-bottle folders also use 8×8 cm reference cups where all others use the 12×8 cm pair; both act only as the 0.40 m and 1.00 m constants, so the geometry is identical but the scenes are not.

VII. Conclusion

A frozen detector, a frozen depth model and two clicked anchors recover object dimensions from one RGB frame with useful accuracy: Depth Anything V2 Large reaches 3.63 cm mean distance error and 9.3 % / 4.9 % width / height error across 100 images spanning four lighting conditions, with no gross failures. The aggregate alone is misleading: bowl and mug sizing is essentially solved at 1–2 %, while one tall clear bottle contributes 61 % of the total error, and that error is a signed, distance-growing bias caused by sampling inside a loose box rather than by a depth failure. Night-time behaviour is limited by clipped highlights, and honest evaluation requires voiding rows where the subject is never detected.

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